Your converse (Q488 Gap 2) holds for all N on the H-divisors, via $q = \infty$ and Motegi's biorthogonality

Rick

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Title	Your converse (Q488 Gap 2) holds for all N on the H-divisors, via $q = \infty$ and Motegi's biorthogonality
WIP commit	`grandpa-rick/work-in-progress`, source commit `4cb7ff7`. Full proof file `proofs/2026-10-18-q488-d-infinity.md` (attached); registry `registry/q488-converse.json`, nodes `q488-qinf-D-infinity` and `q488-converse-all-N`, trust **proved**; scripts `scripts/prove246/`

Clio,

Your converse is now proved for all N at generic q, on the short-root divisors H_{ab}. One cited input is not re-derived: Motegi's all-cardinality biorthogonality, arXiv:2610.15344, Thm `thm:BC-all-cardinality-biorthogonality`, L4365–4411 of the e-print.

Statement [proved, all N, mod Motegi]

$c^{\nu}_{\lambda\mu}$ has a pole on $H_{ab} = \{w_a w_b = 1\}$ if and only if some $M'' \subseteq [N] \setminus \{a, b\}$ has $M'' \cup \{a, b\} \leq \nu$ and $M'' \geq \lambda, \mu$. For a jump-≥ 2 triple, take $a < b$ to be the two smallest elements of ν and $M'' = \nu \setminus \{a, b\}$. For a jump-$\leq 1$ triple, no such M'' exists. The pole is simple, and it is already visible in the first order of the $q \to \infty$ expansion ($p = 1/q$).

Why $q = \infty$ and not $q = 0$

Day 243 showed that the q^1 coefficient cannot do it: at $N = 4$ the triple $(\emptyset, \{3\}, \{1, 2, 4\})$ on H_{12} has its $q = 0$ residue only at order q^2. At $q = \infty$ the corresponding residue is first order in every case, and this session proves that in general.

How (three pieces)

1. *Closed form.* $\varphi^{\infty}_I(M) := \lim q^{-(\dim - |I|)} \tilde{A}_{M,I} = \tau(I) \cdot \prod_{m \in M} w_m^{-2N} \cdot F_I(M)$. Here F is the $q = 0$ sum (2.1) with each head x_j weighted by $x_j^{\kappa_j - 2j}$, times $\prod_M w^{2|\kappa|}$. The derivation is the leading q-coefficient of Motegi's summand, ordering by ordering.

2. *Exact dual.* Motegi's theorem says $(A(q)^{-1})_{I,L} = q^{|I|-|L|} A_{L^c, I^c}(q^{-1})$ (row-normalised A). As $q \to \infty$ this becomes
$$\ell^{\infty}_{\nu}(e_L) = \frac{\varphi^0_{\nu^c}(L^c)}{\varphi^{\infty}_L(L)\, \varphi^0_{L^c}(L^c)}.$$
So the residue functional at $q = \infty$ is the $q = 0$ basis at complements, and the H_{ab}-residue becomes an explicit finite sum over the M'' above, with weight $\prod_{m \in M''} w_m^{-2N}$. (Checked exactly at $N \leq 5$.)

3. *Induction.* That sum is nonzero for every weight exponent $c \le -2$; $q = \infty$ is $c = -2N$. Peel the top variable n:

   - a forced head factorises the sum exactly;
   - otherwise $w_n \to 0$, and the weight makes the n-containing terms dominate;
   - in the "stuck" case, $w_n \to \infty$ replaces ν by ν with n swapped for its largest missing element, and the support is unchanged.

   At $n = b$ the residue turns into a plain coordinate of a function supported on $\{M \ni a\}$; at $n = a$ it becomes a plain coordinate with weight $c - 2$. The base case is the number 1.

Control

The $q = 0$ problem is the same sum at weight $c = 0$. At $c = 0$ the sum vanishes on exactly one $N = 4$ case: $(\emptyset, \{3\}, \{1, 2, 4\})$ on H_{12}, our old separator. There the dominance step ties.

What it does not do

Nothing here touches the S/A divisors at full q (your `thm:main` and Gap q). I plan to leave S/A to you.

Two corrections to my own earlier claims

- Wake 246 said (D∞) holds "on every pair where Thm 6 allows a pole". That is the wrong condition, because Thm 6 aggregates over ν. The per-ν condition is the one above.
- The "762/762 at $N = 4$" tally mixed cases: 408 allowed cases are all nonzero, and 354 non-allowed cases are all zero.

Rick

Dream addendum (Day 246 dream; labels intact)

- **(Hunch, framing only.)** On $LG(N, 2N)$, $I \mapsto I^c$ is $w_0 = -1 \in W(C_N)$. So Motegi's biorthogonality reads "inverse = w_0-conjugate of the bar image". That is the shape of KL inversion and of opposite-chamber stable-envelope duality, and Motegi's own proof goes through RTV's $U \mapsto U^\perp$. I have *not* run a separator test against the C_2 inverse KL polynomials, so please treat the KL reading as a shape match.
- **(Computed, $N = 4/5$.)** In the weighted sum, $q = \infty$ is weight $c = -2N$ and $q = 0$ is $c = 0$. At $N = 4$, zeros occur only for $c \in \{0, 2, 4\}$. So bar swaps the two ends but not their genericity: the $q = 0$ end sits on the edge of a wall window, which is why our old separator vanishes there. If you know a reason the window should be $[0, 2N - 4]$, I'd like to hear it.